

List all 6 phases of SDLC in order with one line about each.

1. **Requirement Analysis** – Gather and analyze the customer's needs and project requirements.
2. **Design** – Create the system architecture, database, and user interface design.
3. **Development (Implementation)** – Write the code and build the software.
4. **Testing** – Test the software to find and fix bugs before release.
5. **Deployment** – Release the software for users to use.
6. **Maintenance** – Update, improve, and fix issues after deployment.

Write 3 differences between Agile and Waterfall models.

1. **Agile** is flexible and allows changes during development, while **Waterfall** follows a fixed sequence of phases.
2. **Agile** delivers the software in small, frequent releases, while **Waterfall** delivers the complete product at the end.
3. **Agile** involves continuous customer feedback, while **Waterfall** has limited customer involvement after requirements are finalized.

Name the 3 main Scrum roles and write one line about the responsibility of each.

- **Product Owner** – Defines the product requirements and prioritizes the work.
- **Scrum Master** – Guides the team, removes obstacles, and ensures Scrum practices are followed.
- **Development Team** – Designs, develops, tests, and delivers the product increment.

What is a Sprint in Scrum? Mention its typical duration.

A **Sprint** is a fixed period during which the Scrum team completes a set of planned tasks and delivers a working product increment.

Typical Duration: 1 to 4 weeks (most commonly **2 weeks**).

What do these Jira board columns mean?

- **Backlog** – A list of tasks or user stories that are planned but not yet started.
- **To Do** – Tasks that are ready to begin.
- **In Progress** – Tasks that are currently being worked on.
- **Done** – Tasks that have been completed successfully.

